LAZYLAYOUT STUDIO | RELEASE 1 · INITIAL PHASE | ROADMAP V4.0.0 | QUALITY ASSURANCE MASTER

LazyLayout: Complete Repository Analysis & Manual Testing Documentation

Comprehensive Architectural Breakdown, Subsystem Analysis, and Phase-by-Phase Quality Assurance Test Execution Manual with Defect Capture Worksheets

A publication-grade testing manual and engineering reference covering the entire LazyLayout codebase: Stage 0 (Foundations 1–12), Stage 1 (Viewport & Layout 13–49), FigCraft Port Deepening (F1–F14), Stage 2 (Animation Engine & Timeline 53–63), cross-browser matrices, chaotic stress suites, and explicit verification for all 50 known issues from AUDIT_ISSUES.md.

202 Phases
SEQUENTIAL ROADMAP ORDER

70+ Completed
FULLY DOCUMENTED TEST TARGETS

216 Properties
CANONICAL MOTION PATHS IN MDM

50 Audit Gaps
EXPLICITLY VERIFIED TEST CASES

Project: LazyLayout (AI-Native Motion Design Studio)

Baseline: Next.js 16 · React 19 · TypeScript 5.x

Date: October 2026 · Confidential QA Guide

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Part I: Comprehensive Repository & Codebase Analysis

Detailed architectural inspection of LazyLayout: product vision, core thesis, technology stack, directory organization, state management, runtime execution engines, security boundaries, and known gaps.

1. Executive Summary & Product Vision

LazyLayout (internally designated *"Unreal Engine for Animation & Frontend Design"* and *"Figma for Motion with AI"*) is an AI-native motion design studio. It bridges the deep gulf between three disciplines that rarely overlap: visual motion design, animation engineering (GSAP, Framer Motion, WAAPI, Three.js, shaders), and production-grade frontend engineering.

In standard industry workflows, designers create animations in Figma or After Effects that cannot be cleanly translated to code without manual rewriting, approximation, and performance degradation. Conversely, frontend developers write complex animation scripts without interactive visual feedback. LazyLayout resolves this by providing:

- **A unified visual canvas** that feels like Figma for authoring responsive layouts, vector shapes, typography, and design tokens.
- **A multi-engine animation timeline** that operates like After Effects (with keyframe twirl-downs, stopwatches, curves, and transport control).
- **A deterministic evaluation kernel** (`evaluate(t)`) guaranteeing that what you scrub and play in the editor matches exported production code down to the sub-pixel (Preview = Export).
- **Multi-target static code generation** compiling to production Next.js 16 (App Router), Vite + React, or standalone HTML with Tailwind, CSS Modules, or vanilla CSS.
- **AI-native authoring** where Claude API or local MCP agents can propose layout and motion changes as typed, reviewable diffs with strict zero-silent-write guarantees.

> **Current Repository State (October 2026):** The repository is operating under **ROADMAP v4.0.0** (202 sequential phases). **Stage 0 (Phases 1–12)** is 100% complete. **Stage 1 (Phases 13–52)** has delivered the complete viewport, drawing tools, inspector, design tokens, components, multi-page routing, and static export (with active FigCraft port deepening F1–F14). **Stage 2 (Phases 53–102)** has merged Phases 53–63 to `main`, delivering schema v6, central clock transport, frame scheduler, WAAPI acceleration, and the Layer Timeline.

2. Core Architectural Laws (The 20 Laws)

Every commit and pull request in this repository is strictly bound by 20 non-negotiable architectural laws:

Law	Name	Architectural Mandate & Invariant
Law 1	**Inside-Out**	Types and schema precede rules; rules precede diagnostics; diagnostics precede UI controls. No UI control may exist without an underlying rule table assertion.
Law 2	**One Document**	Every surface (canvas, inspector, timeline, AI agent, code view, exporter) reads and writes the single Motion Document Model (MDM). No parallel state models.
Law 3	**Preview = Export**	Every animation evaluates through pure mathematics (`evaluate(t)`) such that the live preview and exported code match within 0.5% pixel tolerance.
Law 4	**No Silent AI Writes**	Every AI or agent action generates a visible diff (ghost layers/keyframes) requiring explicit human approval. No silent state modifications.
Law 5	**Real Verification**	Verification gates must run in TypeScript compiler (`tsc`), real browsers (Chromium, Firefox, WebKit), and real production builds, not synthetic mocks.
Law 6	**Simple by Default**	Every capability must expose a Simple face (≤ 7 controls, zero technical jargon) before or alongside its Pro face.
Law 7	**Licensing Gate**	Third-party code cannot be introduced without formal licensing verification in `LICENSES.md` (e.g. GSAP is strictly an export target, never bundled in editor runtime).
Law 8	**One Clock**	A single central transport (`src/core/time/transport.ts`) owns the timeline, playhead, frame rate, and loop mode. No individual panel holds its own timer.
Law 9	**Hot Path**	Per-frame animation values never touch React state or the document store. They flow directly through the FrameScheduler and the DOM node registry.
Law 10	**Document Space**	All layer geometry is stored in document coordinates. Editor UI reads bounding boxes solely through the renderer's measurement API.
Law 11	**One Renderer**	The editor stage and code exporters render from the identical per-archetype render definitions (`src/core/render/definitions/`).
Law 12	**Signal Input Bus**	Continuous inputs (pointer, scroll, touch, keys) are sampled once per frame by the central scheduler. No component attaches unmanaged listeners.
Law 13	**Surface Budget**	Only the GPU compositor creates WebGL/WebGPU contexts within strict hardware device-tier memory limits.
Law 14	**Graceful Degradation**	Every interactive effect defines a fallback chain: GPU Shader → CSS/Canvas → Static Poster → Accessible Reduced-Motion alternative.
Law 15	**Deterministic Replay**	Simulations, particles, and physics replay identically from `(seed, input tape, fixed dt step)` .
Law 16	**Untrusted Code**	User code, code components, and raw shaders run only inside an isolated sandbox with zero access to editor privileges or cookies.
Law 17	**Visual First (No Maths)**	Everything authorable can be created without entering numbers or mathematical formulas (using handles, feel sliders, or visual meters).

Law	Name	Architectural Mandate & Invariant
Law 18	**Sequential Order**	Roadmap phase numbers represent strict execution order. Every phase's dependencies only point backward to lower phase numbers.
Law 19	**Visible Progress**	Every phase from Phase 13 onward must deliver a user-visible change on screen ("You'll see"). At most two infrastructure phases may run consecutively.
Law 20	**Agent Parity**	Every user interaction is a typed command on the command bus, exposed directly to local MCP servers so AI agents can control the entire editor.

3. Technical Stack & Dependencies

Domain	Technology / Library	Role & Configuration
Framework Baseline	Next.js 16 (App Router) + React 19	Strict modern React 19 concurrent features, Server/Client components, and Next.js 16 layout routing.
Language & Types	TypeScript 5.x (Strict Mode)	Full type-safety, zero `any` policy, compiler emitting no errors (`tsc --noEmit` enforced in CI).
Styling & Design System	Vanilla CSS + CSS Modules + TailwindCSS	CSS custom properties design tokens, theme variables, hardware-accelerated transforms, zero ad-hoc CSS.
State Management	Zustand (Decomposed split stores)	Fine-grained stores: `useDocumentStore`, `useEnvironmentStore`, `useProjectStore`, `useSelectionStore`.
Stage Rendering	Isolated `<iframe>` Host (StageHost v2)	Live DOM tree at `/stage-frame.html` with strict CSP `script-src 'none'` and postMessage message-channel dispatch.
Animation Engines	Web Animations API (WAAPI) + Custom Kernel	WAAPI runtime for GPU-accelerated CSS properties; deterministic mathematical kernel (`evaluate()`) for scrub/pause.
3D & Graphics	Three.js 0.186.1 + React Three Fiber + Drei	MIT-licensed WebGL 3D rendering pipeline for meshes, materials, lights, and cameras.
Vector & Booleans	`polygon-clipping` + Kurbo/FigCraft ports	Curve fitting, bezier mathematical evaluations, vector network topology, and boolean clipping operations.
Testing & Parity	Playwright 1.50+ & Node.js Native Test Runner	Cross-browser testing (Chromium, Firefox, WebKit), pixel-parity diffing (pngjs), export build compilation harness.

4. Architectural Subsystems Breakdown

4.1 Unified Motion Document Model (MDM v6)

The core data model lives under `src/core/document/`. It defines 216 canonical property paths organized into a strictly typed document tree:

```
// Conceptual Document Hierarchy
MotionDocument {
  id: string,
  schemaVersion: 6,
  pages: Record<string, Page>,
  layers: Record<string, Layer>,             // Flat dictionary of all layers with parentId links
  compositions: Record<string, Composition>,// Timeline tracks, triggers, work areas, markers
  clips: Record<string, Clip>,               // Keyframe sequences, behaviors, reactive signals
  designComponents: Record<string, Component>, // Component definitions, variants, properties
  tokens: { collections, variables, modes },// Design tokens and semantic color/number variables
  breakpoints: Breakpoint[],                // Responsive breakpoints (Desktop, Tablet, Mobile)
  assets: Record<string, Asset>              // Image/font asset registry with SHA-256 deduplication
}
```

Every property is accessed via dot-notation canonical paths (e.g. `transform.x`, `appearance.opacity`, `paint.fills`, `shape.cornerRadius`, `layout.auto`). All mutations occur through patch-based command transactions that create atomic undo/redo records in IndexedDB.

4.2 Command Bus & Focus Model (Phase 13)

Located in `src/core/commands/`. Every interactive action is a typed command registered with a unique identifier (e.g. `edit.undo`, `canvas.zoomIn`, `transport.play`, `tool.select`). Commands specify VS Code-style `when` context clauses reading from the Zustand context keys store:

- `canvasFocus`: True when pointer hovers the stage canvas.
- `timelineFocus`: True when pointer hovers the Layer Timeline or Sequencer.
- `textEditing`: True when an input, textarea, or canvas inline text editor has focus.
- `selection.count`: Number of currently selected layers.
- `transport.playing`: Whether the animation transport is running.

This completely eliminates keyboard shortcut conflicts (e.g. pressing Space over the timeline plays animation, while pressing Space over canvas activates the Hand panning tool).

4.3 Stage Host v2 & DOM Isolation (Phase 18)

Prior to Phase 18, the editor stage was an HTML string injected into an iframe (`srcDoc`), causing the entire canvas to reload on every keystroke. StageHost v2 mounts a permanent, static iframe (`/stage-frame.html`) governed by strict Content Security Policy (`script-src 'none'`):

- The iframe never runs untrusted scripts or eval.
- Mutations are reconciled incrementally via DOM operations using `LayerNodeRegistry` (Phase 19).
- Editing a layer's color, position, or size updates only that specific DOM node in under 1ms, maintaining smooth 60 FPS even with 500+ complex layers.

4.4 Dual Animation Engine & Parity Verification (Phases 57–60)

LazyLayout solves the classic animation editing paradox (scrubbing requires instant random-access mathematics, while playback requires hardware-accelerated thread independence) using a dual-engine architecture:

- **Scrubbing & Paused State:** The pure TypeScript **Deterministic Evaluation Kernel** (`src/core/kernel/evaluate.ts`) evaluates all property paths at exact time `t`. It integrates GSAP-derived cubic-bezier easings and Framer Motion physical spring formulas.
- **Playing State:** Eligible CSS properties (transforms, opacity, colors) hand off instantly to native browser **Web Animations (WAAPI)** running on the compositor thread.
- **Zero-Pixel Handoff:** Pausing terminates WAAPI and re-applies kernel inline styles with 0.0px jump.
- **Parity Badge:** A real-time drift auditor (`src/core/engine/stage/parity.ts`) samples the stage every 30 frames and computes mathematical variance against `evaluate(t)`. A green badge in the status bar

confirms drift remains under 0.5px.

4.5 FigCraft Port Architecture (F1–F14)

On branch `a/figcraft-port`, LazyLayout incorporates the engine algorithms from the high-performance Rust vector engine FigCraft (MIT licensed), translating data structures and mathematical behaviors into clean TypeScript:

- **Squircle Corner Smoothing (F1):** Continuous-curvature corner radii (G2 continuity) rather than simple circular arcs.
- **Display P3 Color Gamut (F2):** Full wide-gamut color parsing and `color(display-p3 ...)` rendering.
- **Vector Networks & Paint Bucket (F4):** True non-manifold vector topologies with vertices, segments, and faces, editable via the Paint Bucket tool (B).
- **Live Boolean Groups & Shape Builder (F5):** Non-destructive boolean unions/subtractions where operands remain editable layers, plus the interactive Shape Builder tool (Shift+M).
- **Dev Mode (F13):** Inspector showing CSS syntax with token variables, annotations, and kept measurement overlays.

5. Known Audit Issues & Defect Landscape (AUDIT_ISSUES.md)

The repository maintains an active audit log of 50 identified issues discovered during Stage 0–Stage 1 development. Understanding these is essential for thorough manual testing:

ID	Area	Sev	Issue Summary & Defect Behavior
#1	Selection (26)	**Med**	Selection array retains layer IDs that were deleted via undo, redo, or frame removal.
#3	Editing (29)	**Med**	Delete, Cut, Duplicate, Nudge, and Group operations still execute on locked layers if selected via command.
#4	Editing (29)	**High**	Editing shortcuts (Delete, nudge, duplicate, arrange) still fire while Preview mode is active.
#5	Tools (24)	**Med**	Window blur or pointer cancellation mid-drag leaves `gesturing` state active, locking shortcuts.
#7	Document (3)	**High**	Executing undo or redo while a drag transaction is in progress corrupts history and document tree.
#10	Camera (20)	**High**	`NaN` or `Infinity` entering zoom anchor can blank the entire viewport canvas.
#11	Hit-testing (22)	**Med**	Overlapping frames can select a child of the frame underneath, violating visual stacking order.
#18	Transform (27)	**Med**	A pure position move mistakenly writes explicit width/height, converting hug-sized layers into fixed.
#21	Document (3)	**High**	`insertLayers` can overwrite an existing layer if IDs collide without deduplication check.
#23	Clipboard (29)	**High**	Cut command deletes layer from stage even if clipboard write permission was denied.
#24	Time (54)	**High**	Seeking with `NaN` corrupts transport playhead and work area into `NaN`.
#33	Import (3)	**High**	`.lazy.json` import lacks file size and layer count ceilings, allowing hostile memory exhaustion.
#36	Stage (18)	**Med**	`performance.mark` calls executed on every render frame never clear, causing slow memory leak.
#45	Dev Routes	**Med**	Testing lab routes (`/editor/canvas-lab` , `stage-lab` , `render-gallery`) are exposed in production builds.

Part II: Manual Testing Architecture & Execution Framework

Standardized QA protocol, environment initialization, interface topography map, rapid pre-flight smoke testing, and fundamental test recipes.

1. Manual Testing Methodology & The "Do → Expect → Fails if" Standard

To ensure 100% test reproducibility across human testers, automated agents, and multi-browser environments, every test in this guide adheres to the rigorous **Do → Expect → Fails if** tripartite formula:

- **Do:** The exact sequence of physical user interactions—keystrokes, mouse moves, clicks, drags, or menu selections.
- **Expect:** The precise observable visual, temporal, or DOM state that must occur if the implementation is correct.
- **Fails if:** Explicit descriptions of bug states, regressions, timing drift, visual artifacts, or console errors that constitute an immediate test failure.

1.1 Defect Logging & Write-In Protocol

CRITICAL QA INSTRUCTION: Every test case includes a status selector (`Pass` , `Fail` , `Blocked`) and a dedicated ✎ **Actual Outcome / Defect Notes** write-in box. If what you observe matches the **Expect** specification exactly, mark `[X] Pass` . **If the expected outcome was NOT observed, or if unexpected behavior occurred:**

1. Mark `[X] Fail` (or `[X] Blocked` if preceding steps prevent reaching this state).
2. In the provided write-in lines, explicitly record:
    - What actually appeared on screen or failed to appear.
    - The exact text of any Red Console Errors in DevTools.
    - Browser name, version, and operating system.
    - Whether running in `npm run dev` or a production build (`npm run build && npm run start`).

2. Environment Setup & Pre-Flight Configuration

```
# 1. Ensure you are on the correct branch and up to date
cd "/Users/pranav/Project Folder/WebAPPBuilder"
git status

# 2. Install dependencies cleanly (if package-lock changed)
npm ci

# 3. Start the Next.js development server
npm run dev

# 4. Open Chromium (or Brave, Chrome, Edge) to:
http://localhost:3000/editor
```

DevTools Requirement: Open Chrome/Chromium DevTools immediately (`F12` or `Cmd+Option+I`) and navigate to the **Console** tab. Keep it visible throughout testing. **ANY red uncaught exception or error log that appears during an action is an automatic test failure**, even if the user interface appears normal on the surface.

3. Editor Topography Map: Where Everything Is Located

Refer to this interface directory for all button, panel, and shortcut references:

Area	Physical Screen Location	UI Controls, Buttons & Indicators
Application Header	Top bar across full screen width	• **Menus:** File, Edit, View, Window, Help • **Canvas Presets:** Desktop (1440px), Tablet (768px), Mobile (375px) • **Demo Loader:** Mount Showcase Demo button • **Workspace Tabs:** Design workspace, Animate workspace, Code workspace • **Header Tools:** Saved layouts, Open Command Palette (Ctrl+P), Save (Ctrl+S), Undo (Ctrl+Z), Redo (Ctrl+Shift+Z)
Frame Control Bar	Floating strip directly above canvas stage	• Active Frame Name & Dimension boxes (W / H) • Frame Presets dropdown (Desktop, Tablet, Mobile, Custom) • Mode Toggles: **Design** / **Preview** • Snapping: Snap objects, Snap guides, Pixel grid, Rulers (Shift+R) • Canvas Backdrop: Light, Dark, Checkerboard
Tool Dock	Centered horizontally at bottom of stage	• Selection & Hand tools (V, H) • Creation tools: Frame (F), Rectangle (R), Ellipse (O), Line (L), Arrow (Shift+L), Polygon (Y), Star (Shift+Y), Text (T) • Vector & Editing tools: Pen (P), Pencil (Shift+P), Paint Bucket (B), Shape Builder (Shift+M) • Zoom controls: Zoom Out, Current Zoom (Click to 100%), Zoom In, Fit to Screen (Shift+1)
Left Dock	Left vertical sidebar	• **Layers tab:** Hierarchical DOM tree, search input, eye (hide/show), padlock (lock/unlock), motion badges • **Assets tab:** Image/video asset gallery, SVG import, usage counts • **Pages tab:** Multi-page route list (/, /about), add page, section anchors
Right Dock (Inspector)	Right vertical sidebar	• Inspector Tabs: **Design**, **Motion**, **Code** • Design Inspector toggle: **Simple** face vs **Pro** face • Alignment Bar: Align Left, Center, Right, Top, Middle, Bottom, Distribute, Tidy • Property groups: Position, Dimensions, Auto-layout, Constraints, Fill, Stroke, Effects, Typography, Tokens
Bottom Drawer	Bottom section (active in Animate workspace)	• **Layer Timeline tab:** Layer rows, In/Out time bars, Work area (green bar), Markers, Solo/Shy/Lock switches, Property twirl-downs (▸P), Keyframe stopwatches (◷) • **Motion Sequencer tab:** Track-based keyframe editor, playhead needle, speed pills • **Export Preview tab:** Live code and markup generation
Status Bar	Very bottom strip of window	• Layer count readout (e.g. "3 layers selected") • Central Transport Bar: Stop & Return, Step Back (1 frame), Play / Pause, Step Forward, Loop toggle, Timecode (MM:SS:FF) • Parity Badge: `Parity idle` (grey), `Parity OK` (green), `Parity drift` (red)

4. Fundamental Test Recipes

Most tests in this guide build upon two standardized setup recipes:

Recipe A: Make an Animated Demo Page (Motion & Timeline Tests)

1. Open `http://localhost:3000/editor`.
2. Click the **Animate workspace** tab in the header. The bottom drawer opens on **Layer Timeline**.
3. Click the **Motion Sequencer** tab next to it and wait 1 second for the sequencer to hydrate.
4. Click **Mount Showcase Demo** in the header. A complete demo page with a heading and a button appears on the stage.
5. In the Sequencer, click on the ruler at approximately **0.4s**. The playhead jumps there.
6. Click **+ Keyframe**. Two tracks appear: `appearance.opacity` and `transform.y`.
7. Click **Stop and return to start** in the status bar transport (timecode resets to `00:00:00`).
8. Click **Play** in the status bar.

> **Recipe A Success Baseline:** The demo page smoothly fades in and slides upward over ~0.4s, then remains stationary. Status bar timecode counts up continuously.

Recipe B: Make Visual Shapes & Layers (Layout & Editing Tests)

1. Open `http://localhost:3000/editor`.
2. Click **Mount Showcase Demo**.
3. Click the **Rectangle Tool** (dock or shortcut `R`) and drag a box inside the page.
4. Click the **Ellipse Tool** (shortcut `O`) and drag an oval beside it.
5. Click the **Text Tool** (shortcut `T`), drag a text box, and type `Test Run`.

> **Recipe B Success Baseline:** Three new distinct layers are created inside the demo page container, immediately visible in the Layers panel on the left and selectable with the Select tool (`V`).

5. Quick Smoke Test Suite (3-Minute Pre-Flight)

Perform this rapid smoke check before undertaking any detailed phase testing. If any step fails, halt testing immediately and file an environment blocker.

Smoke Test S1 · Editor Boot & Clean Console ☐ Pass ☐ Fail ☐ Blocked

Do: Navigate to `http://localhost:3000/editor` in Chromium with DevTools Console open.

Expect: Editor loads completely. URL redirects to `/editor?projectId=...`. Design workspace tab is active. ZERO red error messages in the console.

Fails if: Blank screen, spinning loader never resolves, 500 server error, or any red uncaught exception appears in console.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Smoke Test S2 · Showcase Demo Mounting ☐ Pass ☐ Fail ☐ Blocked

Do: Click **Mount Showcase Demo** in the top header.

Expect: A white desktop frame mounts on the dotted canvas containing a styled heading and an interactive button. Layers panel populates with the hierarchy.

Fails if: Nothing appears on canvas, canvas flickers to black, or mounting takes longer than 1 second.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Smoke Test S3 · Top Header Menus ☐ Pass ☐ Fail ☐ Blocked

Do: Click each top menu item in sequence: **File**, **Edit**, **View**, **Window**, **Help**. Click canvas to dismiss.

Expect: Each dropdown menu opens instantly with formatted shortcut badges. Clicking outside dismisses cleanly without leaving orphaned popups.

Fails if: Menu fails to open, opens behind the stage iframe, or throws console errors.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Smoke Test S4 · Command Palette (Ctrl+P) ☐ Pass ☐ Fail ☐ Blocked

Do: Press **Ctrl+P** (Cmd+P on Mac). Type `zoom`, then press **Escape**.

Expect: Command palette modal overlays the center of the screen with a focused search input. Typing filters matching commands (e.g. Zoom In, Zoom Out). Escape closes it immediately.

Fails if: Native browser print dialog opens instead of command palette, palette fails to open, or search input cannot be typed into.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Smoke Test S5 · Shape Creation & Undo Transaction ☐ Pass ☐ Fail ☐ Blocked

Do: Select Rectangle Tool (shortcut `R`), drag a box inside the demo frame. Press **Ctrl+Z**. Press **Ctrl+Shift+Z**.

Expect: Rectangle appears on drag release. Ctrl+Z completely removes it in one atomic step. Ctrl+Shift+Z (redo) restores it exactly in place.

Fails if: Shape does not appear, requires multiple undo presses to remove, or undo corrupts the document.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Part III: Manual Testing Suite — Stage 0: Foundation

Verification tests for core architectural infrastructure: Unified Document Model (MDM), transactional history, IndexedDB persistence, rule table, deterministic evaluation kernel, and store decomposition.

Phase 1 · Core Foundation · ✅ CI Green

Build Health, Tooling & Verification Pipeline

What was completed: Strict TypeScript compiler zero-error configuration, ESLint warning budget enforcement (<463 warnings), Next.js 16 / React 19 build validation, Playwright three-browser harness integration, and architectural link checkers.

T1.1 (Dev) · Production Build & Lint Compliance

☐ Pass ☐ Fail ☐ Blocked

Do: Run `npm run typecheck && npm run lint && npm run build` in terminal.

Expect: All three exit with code 0. Zero TypeScript errors. ESLint reports 0 errors and warning budget is respected. Next.js production build completes.

Fails if: Any command exits non-zero, type errors occur, or ESLint budget is exceeded.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phases 2 & 7 · Core Document · ✅ CI Green

Unified Motion Document Model & Canonical Property Paths

What was completed: MDM schema v5/v6 data structures, 216 canonical dot-notation property paths registered in `properties.ts`, document geometry hierarchy, and schema version migrations.

T2.1 (Dev) · Property Path Registry Integrity

☐ Pass ☐ Fail ☐ Blocked

Do: Run `npx tsx --test "src/core/document/__tests__/properties.test.ts"`.

Expect: All property paths resolve correctly with expected interpolation types (number, color, unit, matrix). Zero missing paths.

Fails if: Any test fails, or unregistered property paths are detected.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 3 | Persistence | ✅ CI Green

History Engine, IndexedDB Autosave & .lazy.json Project Files

What was completed: Gesture transaction batching (a 60-frame drag = 1 undo step), IndexedDB non-blocking autosave, dirty state indicator dot, crash recovery, and full JSON export/import of `.lazy.json` project bundles.

T3.1 (User) · Autosave, Dirty Indicator & Reload Recovery

☐ Pass ☐ Fail ☐ Blocked

Do: Mount Showcase Demo. Draw a rectangle. Observe the top Save button. Wait 3 seconds, then hard reload the browser (`Cmd+R` / `F5`).

Expect: Drawing the shape causes a small dirty dot to appear on the Save button. Within ~1s, autosave flushes to IndexedDB and the dot clears. After reload, the rectangle and demo restore identically.

Fails if: Dirty dot remains stuck indefinitely, data is lost on reload, or editor crashes with IndexedDB error.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

T3.2 (User) · .lazy.json Project File Export & Re-Import

☐ Pass ☐ Fail ☐ Blocked

Do: Choose **File → Export Project File (.lazy.json)**. Open the downloaded file in a text editor. In the editor, navigate to `/editor` (blank canvas) and choose **File → Import Project File...**, selecting the exported file.

Expect: The exported file is valid JSON with `schemaVersion: 6`. The re-imported project loads flawlessly with all shapes, text, and properties intact.

Fails if: File download fails, JSON is corrupted, or import throws schema validation errors.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 10 | Rules Engine | ✅ CI Green

Executable Motion Rules Engine & Scored Engine Routing

What was completed: Executable rules compiled directly from the grammar matrix (432 assertions), conditional gating (⚠️ conditional / 🔒 subsumed), accessibility strobe limiter (≤3 flashes/sec), and scored engine routing (CSS vs WAAPI vs Three.js; GSAP excluded from editor runtime).

T10.1 (Dev) · Rule Table Assertions & Routing Score ☐ Pass ☐ Fail ☐ Blocked

Do: Run `npx tsx --test "src/core/rules/__tests__/*.test.ts"`.

Expect: All 432 grammar compatibility assertions pass. `rules.canAdd`, `rules.validate`, and `rules.route` return correct engine choices without importing GSAP.

Fails if: Any matrix assertion fails or illegal engine routing is suggested.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 11 Kernel ✅ CI Green

Deterministic Evaluation Kernel (evaluate())

What was completed: Pure TypeScript mathematical oracle `evaluate(document, time)`. Features exact GSAP cubic-bezier easing solver (Newton-Raphson + bisection), Motion physical spring integrator with velocity handoff, and OKLab color space interpolation.

T11.1 (Dev) · Golden Easing & Spring Determinism ☐ Pass ☐ Fail ☐ Blocked

Do: Run `npx tsx --test "src/core/kernel/__tests__/*.test.ts"`.

Expect: 111 assertions pass across 5 test suites. Kernel easing matches real GSAP golden values; spring solver matches Motion golden values; 200-sample determinism fuzz test passes with 0 drift.

Fails if: Mathematical divergence from golden reference values or non-deterministic outputs.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 12 State Decomposition ✅ CI Green

Store Decomposition & Fine-Grained Subscriptions

What was completed: Decomposed monolithic 1,554-line store into atomic Zustand stores: `useDocumentStore` (document state & patches), `useEnvironmentStore` (camera, grid, lighting), `useProjectStore` (metadata), and `useSelectionStore` (active IDs).

T12.1 (Dev) · Fine-Grained Subscription Performance ☐ Pass ☐ Fail ☐ Blocked

Do: Run `npx tsx --test "src/core/store/__tests__/documentStore.test.ts"`.

Expect: Fine-grained per-clip and per-layer selectors fire only when their respective entity changes. Updates to Layer A do not trigger re-render in Layer B subscribers.

Fails if: Unnecessary subscriber notification cycles or memory retention leaks.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Part III (Cont.): Stage 1 — Viewport, Stage Host, Tools & Selection (Phases 13–28)

Rigorous manual test specifications for command bus routing, dock workspace persistence, security boundaries, stage host iframe isolation, camera navigation, shape creation, spatial hit-testing, selection mechanics, and transform handles.

Phase 13 | Lane A · Viewport | ✅ CI Green

Command Bus, Menu Registry & Focus-Disambiguated Shortcuts

What was completed: Single centralized command registry with context keys (`canvasFocus` , `timelineFocus` , `textEditing`). File/Edit/View/Window/Help menus route commands through the dispatcher. Eliminates conflicting key listeners.

T13.1 · Menu Undo/Redo & Keybinding Dispatch

☐ Pass ☐ Fail ☐ Blocked

Do: Mount demo. Draw a rectangle. Choose **Edit → Undo** from the top menu. Choose **Edit → Redo**. Press **Ctrl+Z** then **Ctrl+Shift+Z**.

Expect: Menu Undo removes rectangle; Menu Redo restores it. Keystrokes produce identical single-step transitions.

Fails if: Menu click does not fire undo, requires double click, or causes desynchronization.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

T13.7 · Disabled Menu States & Pending Phasing

☐ Pass ☐ Fail ☐ Blocked

Do: Open **File** menu. Inspect disabled items: **New Project...**, **Save As Template...**, **Export Next.js + Prisma Code**. Click a disabled item.

Expect: Items are visibly grayed out with phase annotations (e.g. "Phase 51", "Phase 49"). Clicking them produces zero action and ZERO console errors. No fake recent-project items exist.

Fails if: Clicking disabled items triggers crashes or uncaught console exceptions.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 14 · Lane A · Workspace · ✅ CI Green

Workspace Dock Model & On-Demand Panel Registry

What was completed: Panel registry with lazy-loaded chunks. Resizable splitters between side docks and stage. Splitter dimensions and docked panel layout persist across page reloads in `localStorage`. Corrupt layout self-healing.

T14.1 · Splitter Resize & Reload Persistence

☐ Pass ☐ Fail ☐ Blocked

Do: Click **Animate workspace** tab. Drag the vertical splitter before the right dock to widen it significantly. Reload the browser page.

Expect: Page reloads directly into Animate workspace with the right dock retaining the exact widened pixel width.

Fails if: Dock snaps back to default width or resets to Design workspace.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 15 · Lane A · Presets · ✅ CI Green

Workspace Presets (Design, Animate, Code) & Saved Layouts

What was completed: One-click preset switching: Design (Figma-style: stage + inspector), Animate (After Effects-style: timeline drawer + motion inspector), Code (developer-style: live code inspector). Custom named layouts saving and deletion.

T15.1 · Rapid Workspace Switching & Stage Preservation

☐ Pass ☐ Fail ☐ Blocked

Do: Draw a rectangle on the canvas. Rapidly click between **Design**, **Animate**, and **Code** workspace tabs 20 times.

Expect: Switches are instantaneous (<100ms). The center stage frame is NEVER destroyed or reloaded. Drawn rectangle remains stable on screen throughout.

Fails if: Stage flickers to white, iframe reloads, drawn shapes disappear, or editor freezes.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 16 · Lane A · Security · ✅ CI Green

Threat Model & Hostile Content XSS Neutralization

What was completed: Verification of nine trust boundaries and ten security rules (R1–R10). Text layers, layer names, SVG attributes, and AI co-pilot prompts are strictly sanitized against script injection and prototype pollution.

T16.1 · Hostile XSS Injection in Text Layer ☐ Pass ☐ Fail ☐ Blocked

Do: Pick Text Tool (`T`), drag a box, and enter: `<img src=x onerror=alert(1)>`. Double-click layer in Layers panel and rename it to: `<script>alert(2)</script>`.

Expect: **ZERO alert boxes appear.** Both strings render strictly as inert plain text in both the canvas DOM and Layers panel tree.

Fails if: Any browser alert dialog opens, script executes, or DOM elements are injected into editor shell.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 17 · Lane A · Render · ✅ CI Green

Archetype Render Definitions & Render Gallery

What was completed: Replaced monolithic renderer with modular per-archetype definitions (`src/core/render/definitions/`). 53 visual gallery cases covering basic shapes, complex SVGs, badges, floating buttons, and edge cases.

T17.1 · Render Gallery Inspection (53 Cases) ☐ Pass ☐ Fail ☐ Blocked

Do: Open `http://localhost:3000/editor/render-gallery` in browser. Scroll through all cases.

Expect: All 53 cases render with visual clarity (rectangles, ellipses, stars, arrows, paths, dividers, badges). Zero console errors. Malformed inputs stay confined to boxes.

Fails if: Blank test cells, broken images, unstyled fallbacks, or console errors appear.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 18 · Lane A · Stage Host · ✅ CI Green

Stage Host v2: Isolated Live Frame & Strict CSP

What was completed: Retired HTML-string iframe (`srcDoc`). Stage runs inside `/stage-frame.html` with `Content-Security-Policy: script-src 'none'`. Canvas editing modifies the live DOM directly.

T18.1 · Live DOM Frame Selection & Shortcut Pass-Through ☐ Pass ☐ Fail ☐ Blocked

Do: Mount Showcase Demo. Click the heading directly on the canvas. With pointer hovering over the stage, press **Ctrl+Z**, **V**, **H**.

Expect: Heading gets blue outline with dimension badges. Shortcuts fire immediately without clicking parent window to regain focus. Stage does not reload or flicker.

Fails if: Keystrokes are swallowed by the iframe, stage flickers, or selection fails to highlight.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 19 · Lane A · Incremental · ✅ CI Green

Incremental Reconciliation & Measurement API at Scale

What was completed: `LayerNodeRegistry` managing direct DOM element updates. Single property edits bypass React reconciliation. 500-layer stress laboratory (`/editor/stage-lab`) mounting in <500ms.

T19.1 · 500-Layer Mount & Single Node Mutation ☐ Pass ☐ Fail ☐ Blocked

Do: Navigate to `http://localhost:3000/editor/stage-lab`. Click **Load 500 layers**. Click **Edit one layer**.

Expect: 500 layers mount with reported time < 500ms. Clicking "Edit one layer" increments counter and updates only layer 250 in under 2ms without re-rendering the other 499 layers.

Fails if: Mounting takes > 1000ms, page freezes, or editing causes full list re-render lag.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 20 · Lane A · Viewport · ✅ CI Green

Viewport Camera (Pan & Zoom 2%–6,400%)

What was completed: Hardware-accelerated canvas pan and zoom. Smooth cursor-anchored zooming (2% to 6,400%), 100% snap, keyboard shortcuts (Shift+1 fit, Shift+2 zoom selection, Shift+0 100%), per-project camera persistence.

T20.1 · Pan, Pointer Zoom & Clamping Limits ☐ Pass ☐ Fail ☐ Blocked

Do: Hold **Space** and drag on canvas. Use **Ctrl+Wheel** (or pinch trackpad) over a corner of the demo. Click Zoom In until capped, then Zoom Out.

Expect: Space+drag pans fluidly. Zoom anchors around mouse pointer, not canvas center. Zoom stops cleanly at 6,400% maximum and 2% minimum. 100% snaps.

Fails if: Camera produces NaN/Infinity, screen blanks, zoom stutters, or limits are exceeded.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 21 Lane A · Multi-Frame ✅ CI Green

Multi-Frame Responsive Canvas

What was completed: Infinite canvas hosting multiple root frames simultaneously (Desktop 1440px, Tablet 768px, Mobile 375px). Frame titles, dimensions, re-ordering, duplication, and backdrop themes (Light, Dark, Checkerboard).

T21.1 · Responsive Frame Spawning & Title Renaming ☐ Pass ☐ Fail ☐ Blocked

Do: Click **Tablet Canvas (768px)**, then **Mobile Canvas (375px)** in the top header. Double-click the Mobile frame title and rename to `Phone View`.

Expect: Tablet and Mobile frames spawn cleanly beside Desktop frame. Renaming updates title cleanly on canvas and in Layers panel.

Fails if: Frames overlap, spawn off-screen, or renaming throws exception.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 22 Lane A · Hit-Testing ✅ CI Green

Hit-Testing & Spatial Indexing

What was completed: R-Tree/BBox spatial index (`HitIndex`) for sub-millisecond layer hit-testing. Hover outlines, direct child selection vs `Ctrl-click` (Cmd-click) deep selection, and lock/hide safety guards.

T22.1 · Direct Child vs Deep Ctrl-Click Selection ☐ Pass ☐ Fail ☐ Blocked

Do: Mount demo. Click the button inside the demo card. Deselect. Now hold **Ctrl** (Cmd on Mac) and click the text label inside the button.

Expect: Normal click selects the top-level container or direct child. Ctrl-click deep-selects the exact innermost text span layer.

Fails if: Ctrl-click behaves identically to normal click, or selection picks incorrect sibling.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 23 | Lane A · Modes | ✅ CI Green

Overlay Layer & Design vs Preview Modes

What was completed: SVG overlay plane rendering selection rings, hover outlines, and dimension labels. **Preview Mode** toggle rendering live interactive links and buttons while disabling selection handles and edit shortcuts.

T23.1 · Preview Mode Live Interactivity ☐ Pass ☐ Fail ☐ Blocked

Do: Mount demo. Click **Preview** in the frame bar. Hover and click the demo button. Press **Delete**. Click **Design** to return.

Expect: In Preview mode, no outlines or handles appear; button shows CSS hover/active effects; pressing Delete does NOT delete the layer. Switching to Design restores selection.

Fails if: Handles remain visible in Preview, or Delete shortcut deletes layers while previewing.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phases 24 & 25 | Lane A · Tools | ✅ CI Green

Tool State Machine & Shape Creation Tools

What was completed: Modal tool state machine: Select (V), Hand (H), Frame (F), Rectangle (R), Ellipse (O), Line (L), Arrow (Shift+L), Polygon (Y), Star (Shift+Y), Text (T). Drag-to-create with Shift (1:1 ratio / 45° snap) and Alt (center draw).

T25.1 · Shift-Square & Center-Alt Creation ☐ Pass ☐ Fail ☐ Blocked

Do: Press **R**. Hold **Shift** and drag inside frame. Press **O**. Hold **Alt** and drag inside frame.

Expect: Shift forces strict 1:1 aspect ratio (perfect square). Alt draws ellipse expanding symmetrically from the initial click center.

Fails if: Shift allows non-square dimensions or Alt does not expand from center.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 26 | Lane A · Selection | ✅ CI Green

Multi-Selection, Marquee & Hierarchy Navigation

What was completed: Shift-click multi-selection, rubber-band marquee selection across multiple layers, **Ctrl+A** (select all), **Escape** (select none), **Ctrl+Shift+A** (invert), **Enter** (select children), **Shift+Enter** (select parent).

T26.1 · Marquee Drag & Hierarchy Tree Traversal ☐ Pass ☐ Fail ☐ Blocked

Do: Draw two shapes. Drag a marquee over both with Select tool. Status bar reads "2 layers selected". Select the parent frame and press **Enter**. Press **Shift+Enter**.

Expect: Marquee selects both shapes. Enter jumps selection down to children. Shift+Enter moves selection up to the parent frame.

Fails if: Marquee misses intersecting shapes or hierarchy keys fail to navigate.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 27 | Lane A · Transform | ✅ CI Green

Transform Handles, Rotation & Shape Gizmos

What was completed: 8 bounding box transform handles, rotational zones outside corners (Shift snaps to 15°), multi-selection collective transforms, corner radius live handles, polygon sides handle, star points handle.

T27.1 · Corner Resize, Rotation & Radius Gizmo ☐ Pass ☐ Fail ☐ Blocked

Do: Select a rectangle. Drag a corner handle to resize. Hover just outside corner until rotate cursor appears and drag with Shift held. Drag the circular corner-radius handle inside corner. Press **Ctrl+Z** three times.

Expect: Shape resizes smoothly. Rotation snaps in exact 15° increments. Radius handle rounds corners visually. Each gesture undoes cleanly in exactly one step.

Fails if: Rotation angle drifts, radius handle does not round corners, or undo creates intermediate artifacts.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 28 | Lane A · Guides | ✅ CI Green

Snapping, Smart Guides & Canvas Rulers

What was completed: Smart magnetic snapping to edges and centers of adjacent layers. Equal spacing indicators. Alt-hover distance measurement badges. Rulers (Shift+R) with drag-out persistent guide lines.

T28.1 · Smart Guide Lines & Ruler Guide Creation ☐ Pass ☐ Fail ☐ Blocked

Do: Drag a rectangle near another. Press **Shift+R** to show rulers. Drag cursor out of top ruler into canvas. Reload page.

Expect: Red magnetic snapping lines appear when edges align. A horizontal blue guide line is created from ruler. Guide lines remain after reload.

Fails if: Snapping lines fail to display, guide cannot be dragged, or guides vanish on reload.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Part III (Cont.): Stage 1 — Editing, Layers, Inspectors, Paint & Assets (Phases 29–38)

Detailed manual test specifications for Figma-style editing operations, alignment tools, inline text editing, hierarchical layers tree, Simple and Pro inspector faces, paint/gradients engine, asset management, and typography.

Phase 29 | Lane A · Editing | ✅ CI Green

Editing Operations (Clipboard, Duplication, Z-Order, Groups)

What was completed: Cross-tab Copy/Paste (Ctrl+C, Ctrl+V), Cut (Ctrl+X), Duplicate with relative offset memory (Ctrl+D), Arrow key nudging (1px, Shift+arrow 10px), Z-order stacking (Ctrl+[, Ctrl+], Ctrl+Shift+[, Ctrl+Shift+]), Grouping (Ctrl+G, Ctrl+Shift+G), Lock (Ctrl+Shift+L), Hide (Ctrl+Shift+H), and canvas context menu.

T29.1 · Duplicate Repeat Offset (Ctrl+D) & Cross-Tab Paste

☐ Pass ☐ Fail ☐ Blocked

Do: Select a shape. Press **Ctrl+D**. Nudge the copy 20px right and 20px down. Press **Ctrl+D** again. Press **Ctrl+C**, switch to a second tab on same project, and press **Ctrl+V**.

Expect: Second Ctrl+D duplicates again with identical 20px/20px offset applied automatically. Pasting into the second tab pastes the shape identically into that document.

Fails if: Duplicate spawns at 0,0, offset memory fails, or cross-tab clipboard fails to paste.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

T29.7 · Grouping & Ungrouping (Ctrl+G / Ctrl+Shift+G)

☐ Pass ☐ Fail ☐ Blocked

Do: Select two shapes with Shift-click. Press **Ctrl+G**. Verify Layers panel. Press **Ctrl+Shift+G**.

Expect: Ctrl+G wraps both in a new Group container; shapes maintain exact canvas positions. Ctrl+Shift+G ungroups them back to parent container.

Fails if: Shapes shift coordinates when grouped or ungrouping leaves orphan nodes.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 30 | Lane A · Layout | ✅ CI Green

Alignment, Distribution & Grid Tidy Up

What was completed: Top alignment bar and shortcuts: Alt+A (left), Alt+H (horizontal center), Alt+D (right), Alt+W (top), Alt+V (vertical center), Alt+S (bottom). Equal distribution (Ctrl+Alt+H/V), grid Tidy Up (Ctrl+Alt+T), and interactive spacing handles.

T30.1 · Multi-Layer Alignment & Distribution ☐ Pass ☐ Fail ☐ Blocked

Do: Draw three rectangles scattered across frame. Select all three. Click **Align Top** (Alt+W). Click **Distribute Horizontally** (Ctrl+Alt+H).

Expect: Top edges align perfectly to the topmost shape. Horizontal gaps between shapes become exactly equal while outermost shapes remain fixed.

Fails if: Shapes shift out of alignment or distribution calculates unequal spacing.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 31 Lane A · Text ✅ CI Green

Direct Canvas Inline Text Editing & Styled Character Runs

What was completed: Double-click canvas text editing. In-place styled character runs (Ctrl+B bold, Ctrl+I italic, Ctrl+U underline, Ctrl+Shift+X strikethrough on highlighted selection only). Automatic text sizing modes (Auto Width, Auto Height, Fixed).

T31.1 · Inline Formatting & Atomic Undo ☐ Pass ☐ Fail ☐ Blocked

Do: Double-click a text layer on canvas. Type `Alpha Beta Gamma`. Highlight the word `Beta` only and press **Ctrl+B**. Press **Escape** to commit. Press **Ctrl+Z**.

Expect: Only `Beta` becomes bold; `Alpha` and `Gamma` remain normal weight. Pressing Escape commits clean DOM. Ctrl+Z undoes the text edit in a single atomic step.

Fails if: Entire text block bolds instead of selection, or undo takes multiple steps.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 32 Lane A · Layers ✅ CI Green

Layers Panel Tree & Synchronized Selection

What was completed: Visual hierarchical tree in left dock. Drag-and-drop layer reordering and reparenting. Inline rename (F2 / double-click), visibility toggle (eye), lock toggle (padlock), live layer search filter, and context menu.

T32.1 · Canvas ⇄ Layers Panel Bi-Directional Sync ☐ Pass ☐ Fail ☐ Blocked

Do: Select a shape on the canvas. Look at the Layers panel. Select another layer in the Layers panel. Drag a row up in the Layers panel.

Expect: Canvas selection highlights the corresponding row in Layers panel. Panel selection highlights the layer on canvas. Dragging row up reorders DOM stacking order immediately.

Fails if: Selection desynchronizes or layer reordering fails to change visual depth.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phases 33 & 34 | Lane A · Inspector | ✅ CI Green

Design Inspector: Simple vs Pro Faces & Numeric Scrubbers

What was completed: Simple Face (≤7 clean controls: Position, Size, Fill, Stroke, Radius, Opacity; zero jargon) vs Pro Face (hundreds of legal properties, fine-grained transforms, blend modes). Numeric label scrubbing with Shift (10x) and Alt (0.1x) modifiers.

T34.1 · Pro Inspector Label Scrubbing & Undo ☐ Pass ☐ Fail ☐ Blocked

Do: Select a rectangle. Switch inspector to **Pro**. Click and drag horizontally on the **X** label in Position. Observe shape moving. Press **Escape** mid-drag.

Expect: Dragging the label smoothly scrubs the coordinate in real time. Pressing Escape mid-drag cancels the gesture, immediately restoring initial coordinate.

Fails if: Label cannot be scrubbed, values jump wildly, or Escape fails to restore original value.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 35 | Lane A · Paint | ✅ CI Green

Paint Engine: Colors, Gradients, Multi-Strokes, Shadows & Blurs

What was completed: Multi-layer fill stacks. Solid colors (HSB / OKLCH / Hex), linear, radial, and conic gradients with on-canvas drag handles. Multiple strokes, dashed lines, caps/joins, drop shadows, inner shadows, layer blur, and backdrop blur.

T35.1 · Linear Gradient On-Canvas Interactive Gizmo ☐ Pass ☐ Fail ☐ Blocked

Do: Select a rectangle. In Fill section, add a **Linear Gradient**. On canvas, drag the start and end circle handles across the shape. Press **Shift** while dragging end handle.

Expect: Interactive gradient line appears over shape. Dragging handles changes angle and length live. Shift snaps angle in 15° increments.

Fails if: Handles fail to appear, dragging does not update canvas gradient, or Shift snap fails.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phases 36 & 37 | Lane A · Assets | ✅ CI Green

Asset Store, Storage Safety & Interactive Image Fills

What was completed: Drag-and-drop picture upload directly to canvas. SHA-256 deduplication (same file dropped twice creates 1 asset). Script stripping on imported SVGs. Image fill modes (Fill, Fit, Crop, Tile) with interactive on-canvas crop bounding box.

T37.1 · Image Fill Modes & On-Canvas Interactive Crop ☐ Pass ☐ Fail ☐ Blocked

Do: Apply an image fill to a rectangle. Test **Fill**, **Fit**, and **Crop** modes. In Crop mode, drag corner crop handles and middle move handle.

Expect: Fill covers bounds; Fit contains without clipping; Crop exposes on-canvas cropping handles allowing pan/zoom of the image within the container mask.

Fails if: Image distorts aspect ratio improperly or crop gizmo fails to respond.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 38 | Lane A · Typography | ✅ CI Green

Typography, Google Fonts & Custom Web Font Upload

What was completed: Dynamic Google Fonts search and on-demand font fetching (fonts downloaded only when a layer uses them). Custom font upload (.ttf, .otf, .woff, .woff2) embedded into project bundle. Variable font axis sliders (weight, optical size, slant).

T38.1 · Google Font Picker & Dynamic Fetching ☐ Pass ☐ Fail ☐ Blocked

Do: Select a text layer. In inspector typography section, open font family dropdown. Search `Playfair Display` and select it. Open DevTools Network tab.

Expect: Text renders instantly in Playfair Display serif. Network tab confirms font file is requested only upon selection. Selection bounding box automatically adapts to new font glyph metrics.

Fails if: Font falls back to default sans-serif, selection box does not resize, or console throws font fetch error.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Part III (Cont.): Stage 1 — Vector, Layout, Components & Export (Phases 39–49)

Comprehensive manual test specifications for vector pen/pencil authoring, auto-layout flexbox stacks, CSS grid, responsive breakpoint cascades, design tokens/variables, components/variants, multi-page routing, and static code generation.

Phases 39 & 40 | Lane A · Vector | ✅ CI Green

Pen & Pencil Drawing, Path Node Editing & Booleans

What was completed: Pencil tool (Shift+P) gesture curve-fitting algorithm. Pen tool (P) placing nodes and pulling bezier handles (Alt breaks handle tangent). Node editing mode (Enter / double-click), node addition/deletion. Boolean union, subtract, intersect, exclude, and Flatten.

T39.1 · Pen Tool Bezier Curve Creation & Closing

☐ Pass ☐ Fail ☐ Blocked

Do: Press **P** (Pen Tool). Click to place node 1. Drag to place node 2 and pull bezier handles. Click on node 1 to close path. Press **Escape**. Press **Ctrl+Z**.

Expect: Curved path creates cleanly with smooth bezier curvature. Path closes and fills. Entire path creation is reversed in exactly one Ctrl+Z step.

Fails if: Handles fail to pull, clicking node 1 does not close path, or undo takes multiple steps.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

T40.1 · Boolean Union & Destructive Flatten

☐ Pass ☐ Fail ☐ Blocked

Do: Draw two overlapping rectangles. Select both. Click **Boolean Union**. Observe single outline. Click **Flatten to path**. Double-click the resulting shape.

Expect: Shapes merge into a single compound path. Flatten converts it into an editable vector path exposing individual bezier nodes.

Fails if: Boolean produces visual gaps, leaves interior overlapping lines, or fails to flatten.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 41 · Lane A · Auto-Layout · ✅ CI Green

Auto-Layout Stacks (CSS Flexbox with Figma Controls)

What was completed: Convert loose layers into auto-layout flex stack (Shift+A). Direction (horizontal/vertical), wrap, gap, 4-side padding. Child sizing modes: Hug, Fill, Fixed. Drag-and-drop reordering with visual blue insertion indicator line.

T41.1 · Shift+A Stack Creation & Child Drag Reorder

☐ Pass ☐ Fail ☐ Blocked

Do: Draw three buttons side by side. Select all three and press **Shift+A**. In inspector, change gap to `24px`. Drag the first button past the third button.

Expect: Buttons wrap in an auto-layout flex container with 24px gaps. Dragging displays a vertical blue insertion line; releasing drops button into the new sequence position.

Fails if: Buttons overlap, gap does not update, or reorder line fails to display.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 42 · Lane A · Constraints · ✅ CI Green

CSS Grid Layout Containers & Pinned Constraints

What was completed: Grid layout containers with column/row track lists (e.g. `1fr 2fr`, `repeat(3, 1fr)`) and column/row cell spans. Directional constraints for free layers (Left, Right, Left & Right, Center, Scale) adapting when parent frame resizes.

T42.1 · Directional Constraints on Parent Frame Resize

☐ Pass ☐ Fail ☐ Blocked

Do: Draw a rectangle inside a frame. Set its Horizontal Constraint to **Right**. Select the frame and drag its right transform handle outward to widen it.

Expect: The rectangle moves automatically to maintain an exact constant pixel distance from the right edge of the frame.

Fails if: Rectangle remains stationary at absolute X or distorts unexpectedly.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phases 43 & 44 · Lane A · Breakpoints · ✅ CI Green

Responsive Breakpoints Cascade & Multi-View Canvas

What was completed: Responsive Breakpoint Bar (Desktop 1440, Tablet 768, Mobile 375). Downward cascading property overrides. "All" Breakpoint Canvas rendering all device views simultaneously side-by-side with live synchronized updates.

T43.1 · Downward Cascading Overrides (Mobile Edit Isolation) ☐ Pass ☐ Fail ☐ Blocked

Do: Select **Mobile (375px)** in breakpoint bar. Change a heading font size to `18px` and background color to blue. Switch back to **Desktop (1440px)**.

Expect: Desktop retains original large font and original color. Changes made in Mobile are stored strictly as responsive overrides for Mobile only.

Fails if: Editing Mobile modifies the Desktop master styling.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 45 | Lane A · Tokens | ✅ CI Green

Design Tokens, Semantic Variables & Theme Modes

What was completed: Tokens management panel. Semantic color and number variables with multi-mode values (Light / Dark). Binding layer properties (fills, text color, spacing) to tokens. Mode toggle switching whole page instantly. Text and effect styles.

T45.1 · Token Binding & Instant Light/Dark Mode Switch ☐ Pass ☐ Fail ☐ Blocked

Do: Create color variable `brand/primary` (Light: `#3b82f6`, Dark: `#1d4ed8`). Bind two shapes to it. Toggle canvas mode switch from **Light** to **Dark**.

Expect: Both shapes instantly switch fill to the dark mode color. Editing the token's value in the Tokens tab updates all bound layers immediately.

Fails if: Shapes fail to update color or mode toggle does not re-resolve variables.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phases 46 & 47 | Lane A · Components | ✅ CI Green

Components, Instances, Variant Matrices & Container Slots

What was completed: Create Component (Ctrl+Alt+K). Synced instance tree propagation. Instance override isolation with Reset override. Variant properties (e.g. Size: S/M/L, Tone: Primary/Secondary). Container slots for custom instance content.

T46.1 · Main Component Sync & Instance Overrides ☐ Pass ☐ Fail ☐ Blocked

Do: Turn a button frame into a Component (Ctrl+Alt+K). Duplicate it to make an instance. On the instance, override text to `Submit`. On the main component, change background to red.

Expect: Instance background immediately turns red (inheriting main's update), while retaining its custom overridden text `Submit`.

Fails if: Instance fails to inherit color change or main overwrite destroys instance's text override.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 48 · Lane A · Pages · ✅ CI Green

Multi-Page Routing, Sections & Interactive Prototype Links

What was completed: Pages tab in left dock managing routes (`/` , `/about` , `/contact`). Full-width section bands. Shared headers and footers across pages. Layer hyperlink wiring (to page, section anchor, or external URL). Interactive navigation in Preview mode.

T48.1 · Prototype Navigation Link in Preview Mode ☐ Pass ☐ Fail ☐ Blocked

Do: Create an `/about` page. On the home page, select a button and set Link destination to `/about` in inspector. Switch to **Preview** mode and click the button.

Expect: Preview switches instantly to `/about` page. Top address bar updates to `/about` . Back button navigates back to home page.

Fails if: Link click does nothing, page fails to render, or back navigation breaks.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 49 · Lane A · Export · ✅ CI Green

Static Code Exporter (Next.js, Vite, HTML, Tailwind)

What was completed: Export Dialog (Ctrl+E). Compiles entire design to Next.js 16 (App Router), Vite + React, or pure HTML with CSS Modules, Tailwind, or Vanilla CSS. Generates clean component files with typed TypeScript props. ZIP download.

T49.1 · Next.js / Tailwind ZIP Export & Parity ☐ Pass ☐ Fail ☐ Blocked

Do: Mount demo. Choose **File → Export Site...** (Ctrl+E). Select **Next.js (App Router)** and **TailwindCSS**. Click **Export & Download ZIP**. Unpack and inspect generated project.

Expect: ZIP downloads. Contains valid Next.js project structure (`app/page.tsx` , `components/` , `tailwind.config.ts`). Code is clean, dependency-honest, with zero compile errors.

Fails if: Download fails, export contains missing imports, or generated components fail to compile.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Part III (Cont.): FigCraft Port Deepening Suite (Features F1–F14)

Comprehensive manual and developer test specifications for the ported Rust FigCraft engine capabilities: continuous squircle geometry, Display P3 gamut, vector networks, paint bucket, live boolean groups, shape builder, nested component swaps, and Dev Mode.

Feature F1 | FigCraft Port | ✅ CI Green

Shape Geometry: Squircle Corner Smoothing & Arc/Ring Handles

What was completed: Continuous-curvature corner smoothing (squircles, G2 continuity 0–100%) on rectangles; rounded star/polygon vertices; interactive arc gizmos on ellipses (start angle, sweep angle, inner ratio creating donuts/rings).

TF1.1 · Squircle Corner Smoothing & Ellipse Ring Arc Handles

☐ Pass ☐ Fail ☐ Blocked

Do: Select a rectangle. In inspector, set Corner Smoothing to `60%` (iOS squircle). Select an ellipse. Drag the circular Arc Handle on its right perimeter upward, then drag the inner ratio handle inward.

Expect: Rectangle corners transition smoothly into straight edges without tangent discontinuities. Ellipse becomes an open donut ring path governed by `fill-rule="evenodd"`.

Fails if: Corners remain standard circular arcs or arc handles fail to drag.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Features F2 & F3 | FigCraft Port | ✅ CI Green

Display P3 Wide Gamut, Diamond Gradients, Multi-Strokes & Arrowheads

What was completed: Display P3 wide-gamut color parsing and `color(display-p3 ...)` rendering. Exact four-quadrant diamond gradients. Multiple independent strokes per shape with inside/center/outside alignment, per-side weights, and custom arrowheads.

TF3.1 · Stroke Alignment (Inside / Center / Outside) & Arrowheads ☐ Pass ☐ Fail ☐ Blocked

Do: Select a shape. Set a 10px stroke. Cycle alignment: **Inside**, **Center**, **Outside**. Draw a line (`L`) and set start cap to **Arrow** and end cap to **Circle**.

Expect: Stroke shifts position cleanly across boundary. Outside stroke does not intrude into inner fill. Line renders with an arrowhead on start and a solid circle on end.

Fails if: Stroke alignment does not move or line caps render as plain flat ends.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Feature F4 FigCraft Port ✅ CI Green

Vector Networks & Interactive Paint Bucket Tool (B)

What was completed: Non-manifold vector network topology (vertices connecting arbitrary branching edges). Automatic face discovery. Paint Bucket tool (shortcut `B`) clicking inside closed vector regions to fill or unfill enclosed areas.

TF4.1 · Paint Bucket Region Filling & Unfilling (B) ☐ Pass ☐ Fail ☐ Blocked

Do: Draw a rectangle, select it, click **Flatten to path**. Press **B** (Paint Bucket). Click inside the shape. Click inside it again. Press **Ctrl+Z**.

Expect: First click unfills the bounded region (path becomes hollow outline). Second click refills it with the active fill paint. Ctrl+Z undoes the fill state in one step.

Fails if: Paint bucket does not change fill or fails to identify closed network face.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Feature F5 FigCraft Port ✅ CI Green

Live Boolean Groups & Shape Builder Tool (Shift+M)

What was completed: Non-destructive Live Boolean Groups where operands remain individual editable layers. **Shape Builder Tool** (Shift+M) allowing direct click or drag across overlapping shape regions to merge, or Alt-click to delete regions.

TF5.1 · Live Boolean Non-Destructive Operation Switching ☐ Pass ☐ Fail ☐ Blocked

Do: Select two overlapping rectangles. Click **Live boolean group (union)**. Inspect Layers panel. In inspector, switch Boolean operation to **Subtract**. Press **Ctrl+Z**.

Expect: A live boolean group is created; both operands still exist as layers inside it. Switching to Subtract instantly recalculates the visual cutout. Ctrl+Z returns to Union.

Fails if: Operands are permanently destroyed, or switching operation fails to update stage.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

TF5.2 · Shape Builder Overlap Region Merge (Shift+M) ☐ Pass ☐ Fail ☐ Blocked

Do: Draw two overlapping rectangles. Select both. Press **Shift+M** (Shape Builder). Hover over the overlapping center intersection. Click it.

Expect: Hovering highlights the exact intersecting segment. Clicking extracts the intersection into a new distinct shape while trimming the parent rectangles.

Fails if: Shape builder does not detect intersection, or click fails to create region.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Feature F7 FigCraft Port ✅ CI Green

Advanced Effects: Progressive Blurs & SVG Noise Turbulences

What was completed: Multi-stop Progressive Background Blur (blur radius ramps linearly between two spatial points); SVG procedural noise grain filters (mono, duo, full-color) using `feTurbulence` and `feFlood`; simulated frosted glass with highlight reflection.

TF7.1 · SVG Noise Filter & Progressive Blur Bands ☐ Pass ☐ Fail ☐ Blocked

Do: Select a shape. In Pro inspector Effects, add **Noise**. Switch mode to **Duo**. Add a **Progressive Background Blur** with radius 16px.

Expect: Shape displays fine procedural grain texture via SVG filter. Progressive blur splits into 5 backdrop blur bands smoothly ramping optical depth.

Fails if: Noise renders as solid gray or blur does not ramp progressively.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Features F9 & F10 | FigCraft Port | ✅ CI Green

Nested Components, Push Changes & Token Variable Aliases

What was completed: Nested component instance swaps; "Push changes to main component"; Combine as variants; variable token collections with aliases (e.g. `surface → bg`) and circular dependency refusal; frame-scoped collection modes.

TF9.1 · Push Changes to Main & Token Alias Resolution

☐ Pass ☐ Fail ☐ Blocked

Do: Create component. Duplicate an instance. Change instance background color. Click **Push changes to main component** in Instance inspector.

Expect: The modified color pushes backward to the main component, which in turn propagates to all other unmodified instances across the document.

Fails if: Push changes button is disabled or main component fails to adopt the new styling.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Features F13 & F14 | FigCraft Port | ✅ CI Green

Dev Mode Code Inspection & Native Editable SVG Vector Import

What was completed: Dev Mode Code Tab showing generated CSS with token variable syntax (`var(--token)`), annotations, and measurement overlays. Native SVG import parsing compound paths, gradients, and groups into fully editable vector design layers.

TF14.1 · SVG Import to Fully Editable Design Vectors

☐ Pass ☐ Fail ☐ Blocked

Do: In Assets panel, click **Import Design SVG...**. Select an SVG with paths and text. Double-click the imported layers on canvas.

Expect: SVG imports not as a static image, but as a structured frame containing native path, text, and group layers with editable bezier nodes.

Fails if: SVG imports as an inert `<img>` tag or throws SVG parser errors.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Part III (Cont.): Stage 2 — Animation Grammar, Transport & Timeline (Phases 53–63)

Exhaustive manual and developer test specifications for the central transport clock, deterministic frame scheduler, WAAPI acceleration, scrub vs play handoff, real-time parity verification badge, Layer Timeline, and After Effects keybindings.

Phase 53 | Lane B · Schema v6 | ✅ CI Green

Schema v6 Migration: Advanced Clip Types & Triggers

What was completed: Migration from Schema v5 to v6. New clip types (`exit` , `focus` , `stagger` , `layout`) and triggers (`unmount` , `layoutChange`). Automatic backward-compatible migration of older `.lazy.json` project bundles.

T53.1 · Schema Migration from Older Projects ☐ Pass ☐ Fail ☐ Blocked

Do: Import `tests/e2e/fixtures/showcase-v4.lazy.json` . Choose **File → Export Project File (.lazy.json)**. Open in text editor and inspect `schemaVersion` .

Expect: Older project opens flawlessly with zero console warnings. Exported bundle automatically migrates and writes `"schemaVersion": 6` .

Fails if: Import throws migration exception or exported version remains at 4 or 5.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 54 | Lane B · Transport | ✅ CI Green

Central Transport Bar & Single Clock Authority

What was completed: Single clock authority (`src/core/time/transport.ts`) managing time, 60fps frame rate, loop mode (Off, Loop, Ping-Pong), and shuttle speeds. Transport controls in status bar with precise `MM:SS:FF` timecode. Keyboard shuttling (J/K/L).

T54.1 · Play, Pause, Single-Frame Stepping & Ping-Pong Loop ☐ Pass ☐ Fail ☐ Blocked

Do: Click **Play** in status bar. Wait 2 seconds. Click **Pause**. Click **Step forward one frame** 5 times. Click Loop button twice to set **Loop: ping-pong**. Click Play.

Expect: Timecode advances to ~ `00:02:00`. Pause stops timecode dead. Stepping advances exactly 5 frames (`00:02:05`). In ping-pong mode, playback counts up to 5s, reverses smoothly toward 0s, and repeats.

Fails if: Timecode drifts, skips frames, or ping-pong fails to count backwards.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

T54.16 · J / K / L Timeline Shuttling Keys ☐ Pass ☐ Fail ☐ Blocked

Do: Hover mouse over timeline panel. Press **L** (plays 1x). Press **L** again (2x), again (4x). Press **J** (slows to 2x, 1x, pauses). Press **J** from standstill (plays backwards). Press **K** (stops dead). Click into a text box and press **L**.

Expect: L accelerates shuttle speed forward; J reverses shuttle; K stops immediately. Typing in a text box types the letter 'L' and does NOT trigger transport.

Fails if: Shuttle keys fail or fire while typing in text inputs.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phases 55 & 56 | Lane B · Scheduler | ✅ CI Green

Deterministic Frame Scheduler & Sequencer Clock Integration

What was completed: Single `requestAnimationFrame` loop ordering per-frame tasks: `input → evaluate → apply → render → overlay → stats`. Background tab gap clamp (≤0.25s). Motion Sequencer unified onto central transport clock.

T56.1 · Sequencer Ruler Scrubbing & Status Bar Synchronization ☐ Pass ☐ Fail ☐ Blocked

Do: Execute Recipe A. Click at **1.5s** on the Sequencer ruler. Drag the playhead needle handle horizontally back and forth between 0s and 3s.

Expect: The Sequencer needle tracks the mouse with zero lag. Status bar timecode updates simultaneously. Both clocks match within 1 frame (e.g. `00:01.500` = `00:01:30`).

Fails if: Needle lags behind mouse, timecodes disagree, or scrubbing causes visual stutter.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phases 57 & 58 | Lane B · Kernel & WAAPI | ✅ CI Green

Evaluation Kernel on the Stage & WAAPI Hardware Acceleration

What was completed: Kernel output applies live to DOM elements during playback/scrubbing. Eligible simple properties (transforms, opacity, colors) promote to native browser Web Animations API (`ll:<layer_id>`) running on GPU compositor thread.

T57.1 · Live Stage Animation & Playhead Scrubbing

☐ Pass ☐ Fail ☐ Blocked

Do: Execute Recipe A. Click **Stop and return to start**. Slowly drag playhead needle from 0s to 1s. Click **Play**.

Expect: Dragging needle scrubs the demo page opacity and slide-up smoothly in exact real-time sync with mouse. Clicking Play executes silky 60fps GPU playback.

Fails if: Stage elements do not move when scrubbing, or animation jumps abruptly.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phases 59 & 60 | Lane B · Parity | ✅ CI Green

Zero-Pixel Scrub vs Play Handoff & Real-Time Parity Badge

What was completed: Seamless transition between kernel mathematical evaluation (paused/scrubbing) and WAAPI hardware playback. Real-time parity auditor sampling stage every 30 frames and displaying status badge: `Parity idle` (grey), `Parity OK` (green), `Parity drift` (red, >0.5px drift).

T59.1 · Zero-Pixel Pause Handoff & Green Parity Badge

☐ Pass ☐ Fail ☐ Blocked

Do: Execute Recipe A. Start playback with **Play**. Observe parity badge on right side of status bar. While demo is mid-fade, click **Pause**. Step 1 frame forward, then backward.

Expect: Parity badge turns green (**Parity OK**). Clicking Pause produces ZERO visible jump or flicker. Frame stepping smoothly increments/decrements with exact kernel matching.

Fails if: Parity badge turns red, or pausing snaps stage elements to different coordinates.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 61 | Lane B · Layer Timeline | ✅ CI Green

The Layer Timeline (Rows, Time Bars, Work Area & Markers)

What was completed: Full Layer Timeline workspace. One row per layer with time bars. In/Out trim handles, time sliding. Green Work Area bar with dedicated start/end handles. Yellow timeline markers. Layer switches: Solo (S), Visibility (●), Lock (🔒), Shy (~), Motion blur (≈), Blend mode, and Parent menu. Motion badges in Layers panel.

T61.1 · Layer In/Out Trimming & Solo Switch (S) ☐ Pass ☐ Fail ☐ Blocked

Do: In Animate workspace, open **Layer Timeline**. Expand root layer. On heading row, drag the right bar handle inward to 2s. Click **S** (Solo) on heading row. Press **Ctrl+Z**.

Expect: Heading bar trims to 2.0s with white outline. Solo mode hides all other layers on stage except heading. Ctrl+Z cleanly un-solos the layer.

Fails if: Trim handle does not drag, solo fails to hide other layers, or undo fails.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 62 | Lane B · Keyframing | ✅ CI Green

Twirl-Downs, Keyframe Stopwatches (◷) & U Hotkey Filtering

What was completed: Property twirl-downs (▸P) opening groups: Transform, Appearance, Effects, States, Behaviours, Links. Stopwatch icons (◷) toggling keyframing and spawning yellow keyframe diamonds on ruler. **U** hotkey revealing animated properties; **UU** revealing all modified properties.

T62.1 · Stopwatch Keyframe Toggling & U Reveal Hotkey ☐ Pass ☐ Fail ☐ Blocked

Do: In Layer Timeline, click **▸P** on heading row. Expand **Transform**. Move playhead to 1s and click stopwatch ◷ on **Position X**. Select layer name, hover timeline, and press **U**.

Expect: Stopwatch turns bright yellow; yellow diamond appears at 1s. Pressing U hides all un-animated properties, isolating Position X in the row view.

Fails if: Stopwatch does not turn yellow, diamond fails to spawn, or U fails to filter.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Phase 63 | Lane B · Navigation | ✅ CI Green

After Effects Timeline Navigation Hotkeys

What was completed: After Effects industry-standard keybindings active over Layer Timeline: **Home** (comp start), **End** (comp end), **PageDown / PageUp** (step 1 frame), **Shift+PageDown / Up** (step 10 frames), **I** (jump to In-point), **O** (jump to Out-point), **B** (set Work Area Start), **N** (set Work Area End), **= / -** (zoom timeline), **** (zoom to fit Work Area).

T63.1 · I / O In-Out Jumping & B / N Work Area Setting ☐ Pass ☐ Fail ☐ Blocked

Do: Trim heading layer bar to 1s–3s. Select heading. Hover mouse over timeline. Press **I**. Press **B**. Press **O**. Press **N**. Press **** (backslash).

Expect: I jumps needle to 1s; B snaps work area start to 1s. O jumps needle to 3s; N snaps work area end to 3s. Backslash zooms timeline horizontally so the 1s–3s work area fills the visible viewport width.

Fails if: Shortcuts fail to navigate, or keys trigger canvas actions instead.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

Part IV: Regression, Cross-Browser, Stress & Audit Testing

Holistic integration validation: 25-step multi-feature user journeys, cross-browser compatibility matrix (Chromium, Firefox, WebKit), 15 chaotic stress test scenarios, and comprehensive verification suite for the 50 known issues from AUDIT_ISSUES.md.

1. End-to-End Regression Journeys

Execute these complex multi-feature workflows to verify that independent subsystems interoperate seamlessly without state corruption:

Journey J1 · Full Lifecycle: Design → Animate → Inspect → Export ☐ Pass ☐ Fail ☐ Blocked

Do: 1. Boot editor on fresh project.
2. Draw Desktop frame (1440px), add two rectangles and a text layer.
3. Wrap shapes in an auto-layout flex stack (Shift+A) with 20px padding.
4. Convert stack to a Component (Ctrl+Alt+K) and duplicate an instance.
5. Switch to **Animate workspace**. Twirl open instance properties (▸P) and toggle Position X stopwatch at 1s.
6. Scrub playhead and press Play in transport bar (verify parity badge is green).
7. Switch to **Code workspace** and inspect generated CSS.
8. Open File → Export Site (Ctrl+E) and export Next.js + Tailwind project.

Expect: Every transition operates smoothly without lag. The exported site reproduces the exact canvas layout and motion. Parity badge remains green throughout.

Fails if: Any step throws an error, components desync, or export fails.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

2. Cross-Browser Compatibility Matrix

Run these critical checks across **Chromium** (Chrome/Brave/Edge), **Firefox**, and **WebKit** (Safari):

Test ID	Feature Area	Cross-Browser Verification Action	Engine Differences & Known Traps
XB-01	StageHost CSP & PostMessage	Verify stage loads cleanly in WebKit with `script-src 'none'`.	WebKit enforces strict origin policies on iframe communication.
XB-02	Zoom & Pan Canvas 60 FPS	Pan and zoom with 5 frames × 200 layers in Firefox and WebKit.	Check AUD-37 (WebKit zoom frame ceiling with high layer counts).
XB-03	WAAPI Animation Execution	Execute Recipe A and observe Web Animations in Firefox DevTools.	Firefox sub-frame timing precision differs slightly from Blink.
XB-04	Squircle & Arc SVG Rendering	Render Gallery in Safari: verify smooth squircle corners and donut rings.	WebKit requires explicit `fill-rule="evenodd"` on compound paths.
XB-05	Keyboard Focus Routing	Hover canvas and timeline in Firefox; test Space (Hand vs Play).	Firefox pointer-enter dispatch order can differ under rapid mouse sweeps.

3. Chaotic Stress & Break-Testing Scenarios (15 Edge Scenarios)

Perform these aggressive torture tests to verify error recovery, memory safety, and boundary enforcement:

Stress S1–S5 · Input & Memory Torture Suite ☐ Pass ☐ Fail ☐ Blocked

Do:
- **S1:** Rapidly alternate between **L** and **J** 50 times over the Sequencer.
- **S2:** Switch workspaces 30 times as fast as possible.
- **S3:** In stage-lab, mount 500 layers and hold down **Ctrl+Z** continuously.
- **S4:** Leave looping animation running in background for **15 minutes** with DevTools Memory tab recording.
- **S5:** Open the same project simultaneously in two browser tabs and edit shapes in both.

Expect: Editor does not crash. Transport handles shuttling without locking. Memory graph stays completely flat (no leaks). Second browser tab opens safely in read-only mode.

Fails if: Browser tab crashes, memory climbs continuously (leak), or history is corrupted.

✎ ACTUAL OUTCOME / DEFECT NOTES (FILL IF EXPECTED OUTCOME NOT OBSERVED)

4. Comprehensive Audit Issues Verification Matrix (AUDIT_ISSUES.md)

Explicit test cases targeting all 50 known issues documented in `AUDIT_ISSUES.md`:

#	Area	Sev	Targeted Test Action	Expected Pass Behavior	Status
1	Selection (26)	Med	Delete a frame containing 3 selected layers. Inspect `useSelectionStore.ids`.	Deleted layer IDs are immediately purged from selection array.	[ ] Pass [ ] Fail
2	Commands (13)	Low	Select 2 layers. Check context key `selection.count`.	Context key updates accurately to `2`.	[ ] Pass [ ] Fail
3	Editing (29)	Med	Lock a layer. Select it via command and press Delete or Nudge.	Locked layer is strictly protected and does NOT delete or move.	[ ] Pass [ ] Fail
4	Editing (29)	High	Switch to **Preview** mode. Press Delete, Ctrl+D, Ctrl+G.	Editing shortcuts are strictly disabled while Preview mode is active.	[ ] Pass [ ] Fail
5	Tools (24)	Med	Begin dragging a shape, press Alt+Tab to blur window mid-drag.	`gesturing` state terminates safely; shortcuts do not lock up.	[ ] Pass [ ] Fail
6	Tools (25)	Low	Begin dragging a rectangle. Press Escape before releasing mouse.	Shape creation is aborted cleanly; no layer is spawned.	[ ] Pass [ ] Fail
7	Document (3)	High	Press Ctrl+Z while actively dragging a transform handle.	Undo is either ignored during active drag or terminates transaction safely.	[ ] Pass [ ] Fail
8	Document (3)	Med	Execute `jumpTo` to an invalid history state index.	Editor aborts cleanly without entering an infinite loop.	[ ] Pass [ ] Fail
9	Camera (20)	Med	Manually corrupt localStorage camera zoom to `10000%` and reload.	Zoom clamps automatically to legal range (2%–6,400%).	[ ] Pass [ ] Fail
10	Camera (20)	High	Pass `NaN` to `setCamera()` in console.	Camera rejects non-finite values; canvas does not blank.	[ ] Pass [ ] Fail
11	Hit-test (22)	Med	Stack Frame B over Frame A. Click overlapping area.	Click selects child of Frame B (topmost), respecting paint order.	[ ] Pass [ ] Fail
12	Frames (21)	Med	Move a frame vertically and execute `moveFrame`.	Frame respects custom Y position; does not force single-row reset.	[ ] Pass [ ] Fail
13	Frames (21)	Med	Delete a top-level frame. Check `page.rootElementId`.	Page updates root ID safely without pointing to orphaned null.	[ ] Pass [ ] Fail
14	Frames (21)	Med	Create a new empty frame in a blank project. Select Rectangle tool.	Empty frame is rendered and accepts shape drawing immediately.	[ ] Pass [ ] Fail
15	Guides (28)	Low	Attempt to create 500 guides on a frame via script.	System enforces sensible guide limit (e.g. max 50).	[ ] Pass [ ] Fail

#	Area	Sev	Targeted Test Action	Expected Pass Behavior	Status
16	Geometry	Med	Pass non-finite coordinates to `dragBox` and `unionBox`.	Geometry helpers reject `NaN` / `Infinity`.	[ ] Pass [ ] Fail
17	Persistence	Low	Delete a project. Check localStorage for lingering camera keys.	Associated project camera keys are cleanly unlinked.	[ ] Pass [ ] Fail
18	Transform (27)	Med	Move a hug-sized container across the canvas with mouse.	Move updates X/Y only; does NOT overwrite width/height to fixed.	[ ] Pass [ ] Fail
19	Clipboard (29)	Med	Copy 5,000 layers to clipboard and paste.	Clipboard either pastes successfully or reports clear payload limit.	[ ] Pass [ ] Fail
20	Document (2)	Med	Duplicate a document containing 10,000 layers.	`createId` generates non-colliding 64-bit UUIDs.	[ ] Pass [ ] Fail
21	Document (3)	High	Insert a layer with an ID matching an existing layer.	Collision is detected and layer is re-keyed rather than overwritten.	[ ] Pass [ ] Fail
22	Clipboard (29)	Low	Select a locked container and press Ctrl+V.	Paste into locked or hidden containers is rejected with message.	[ ] Pass [ ] Fail
23	Clipboard (29)	High	Trigger Cut with clipboard permission blocked.	Layer is NOT deleted from stage if clipboard copy failed.	[ ] Pass [ ] Fail
24	Time (54)	High	Execute `transport.seek(NaN)` in console.	Seek is rejected; playhead does not corrupt to `NaN`.	[ ] Pass [ ] Fail
25	Time (54)	Med	Define a composition with `fps: 0`.	Composition validator rejects `fps: 0`, defaulting to 60.	[ ] Pass [ ] Fail
26	Time (54)	Med	Set work area start to 4s and end to 2s (inverted).	Work area auto-corrects or clamps to minimum 1-frame duration.	[ ] Pass [ ] Fail
27	Time (54)	Med	Play composition at fractional frame rate (23.976 fps).	Timecode prints cleanly formatted `MM:SS:FF` without float blur.	[ ] Pass [ ] Fail
28	Time (54)	Med	Execute `tick(Infinity)` or advance by `NaN`.	Time validator discards non-finite delta step.	[ ] Pass [ ] Fail
29	Time (54)	Low	Set transport playback rate to `1e12`.	Rate clamps to legal playback range (0.1x to 8x).	[ ] Pass [ ] Fail
30	Time (55)	Low	Register a throwing task in the FrameScheduler.	Scheduler catches error, logs once, and disables task to prevent log flood.	[ ] Pass [ ] Fail
31	Time (55)	Low	Execute `FrameScheduler.step(-1)`.	Negative or `NaN` step is safely rejected.	[ ] Pass [ ] Fail

#	Area	Sev	Targeted Test Action	Expected Pass Behavior	Status
32	Persistence (3)	Med	Simulate IndexedDB failure during autosave.	Autosave schedules automatic retry backoff.	[ ] Pass [ ] Fail
33	Import (3)	High	Attempt to import a 200MB corrupted JSON file.	Import validates file size ceiling before parsing, preventing OOM.	[ ] Pass [ ] Fail
34	Import (3)	Low	Export project with name `CON.lazy.json` or control chars.	File name is sanitized to prevent filesystem reserved clashes.	[ ] Pass [ ] Fail
35	Search (8)	Med	Inject invalid string into localStorage recent searches key.	Search panel catches corrupted data, falls back to empty array.	[ ] Pass [ ] Fail
36	Stage (18)	Med	Perform 1,000 edits in stage-lab. Inspect `performance.getEntries()`.	Performance marks are cleared periodically; zero memory leak.	[ ] Pass [ ] Fail
37	Camera (20)	Med	Pan and zoom in WebKit with 5 frames × 200 layers.	Frame time remains under 16ms (60 FPS) without 40ms spikes.	[ ] Pass [ ] Fail
40	Menus (13)	Low	Inspect top menus for built features (e.g. Rulers, Presets).	Menu items do not display outdated "Coming in Phase 14" text.	[ ] Pass [ ] Fail
43	Commands (13)	Med	Run automated keybinding audit spec.	Zero shortcut clashes detected across all context modes.	[ ] Pass [ ] Fail
44	Canvas (20)	Low	Inspect zoom constants in codebase.	Zoom limits (2% and 6,400%) are centralized in single constants file.	[ ] Pass [ ] Fail
45	Dev Routes	Med	Run `npm run build && npm run start`. Navigate to `/editor/stage-lab`.	Development lab routes are disabled or return 404 in production.	[ ] Pass [ ] Fail
46	Rulers (28)	Low	Change a shape fill color while rulers are displayed.	Rulers do NOT repaint on un-related paint mutations.	[ ] Pass [ ] Fail
47	Hit-test (22)	Low	Drag a shape continuously across 1,000 layers.	Spatial index updates incrementally; drag remains smooth 60 FPS.	[ ] Pass [ ] Fail
48	Editing (29)	Low	Select two layers in different containers and press Ctrl+G.	Group handles cross-container grouping cleanly with reparenting.	[ ] Pass [ ] Fail
49	Frames (21)	Low	Spawn new frames with custom vertical offsets.	`nextFramePosition` places frame relative to visual topmost.	[ ] Pass [ ] Fail
50	Overlay (23)	Low	Hover canvas and press Ctrl without moving mouse.	Hover outline instantly switches to deep-select highlight.	[ ] Pass [ ] Fail

5. Standardized Defect Reporting Guide & Bug Filing Template

When any test case fails, copy this standardized template to report the issue:

````
### [DEFECT REPORT] Phase <Phase_Num> · Test <Test_ID>

- **Severity:** [High (Data Loss / Crash) | Med (Broken Feature) | Low (Cosmetic)]
- **Phase & Area:** Lane <A|B>, Phase <NN>, Subsystem: <e.g. StageHost / Timeline / Paint>
- **Browser & Environment:** <e.g. Chromium 130 / macOS 15.1 / npm run dev>
- **Steps to Reproduce (Do):**
  1.
  2.
  3.
- **Expected Outcome (Expect):**
- **Actual Observed Outcome (Fails if):**
- **Console Output / Uncaught Exceptions:**
  ```
  <paste red console errors here>
  ```
- **Screenshots / Video Timestamp:**
````